

Name: _____

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Class: _____

MyDesign Portfolio

Element G Initial Template

Guidance before filling out the template that follows:

- Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of this Element as another piece of a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element also.
- You will then proceed on to test and analyze this design and then determine your next iteration, returning to this Element afterwards when you are ready to construct a new testable prototype. Fill out each of these steps again.

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

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Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Measure pvc tubing	Ruler, pencil, pvc pipe
Step 2	Cut the pvc piping pieces.	Pvc cutter, pvc pipe
Step 3	Assemble pvc frame	Pvc corner pieces, cut pvc pipe
Step 4	Glue 5 ml face to frame	Ruler, Marker, Wire Cutters, 5 ml Frame, Hot Glue
Step 5	Glue 1 ml around frame into Cone	Hot Glue, Ruler, Marker, Wire Cutter
Step 6	Sew Corners together	Needle and Thread,
Step 7	Glue Can to end of mesh net	Hot Glue

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

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Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Cut PVC and create a rectangular foundation to attach the cone + holder.	PVC cutter & PVC pieces
Step 2	Cut out 5mm netting to cover the rectangular foundation, and create a cone with 1mm netting	Cutting tools, 5mm net, 1mm net.
Step 3	Attach the 1mm cone net to the rectangular base.	Hot glue
Step 4	Attached the cone to the pyramid cone	Hot glue
Step 5		
Step 6		
Step 7		

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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
PVC	4-way connector	Formufit - Amazon	1.583	12	19
PVC	Tubing	Unsure	Unsure	11	Unsure
PVC	Two-way connectors	Unsure	Unsure	8	unsure

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective,

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measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Able to fully submerge	Because it can get fully under the water, it contributes to how it will collect microplastics.
Manuverable	Able to navigate as low as the bay floor was deep, around 6ft (next to a doc), also able to go as far as the tether could allow, 15 ft+.
Can properly collect microplastics	Being able to collect the required amount of microplastics, which is allowed around 5ml, with possible sample sizes of 5mm

Testing or Modeling of Attributes (*G3, G4*) and Addressing Non-Testable Attributes (*G4*)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

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For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter “expert review required” for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute
Motors	3 motors allow for the required movement of going up and down, as well as left or right. Positioned with one in the back and one on each side of the ROV. Takes charge of navigation of the bay and not it's able to submerge and return to the surface.	Was able to run perfectly in tests, both in and out of water.
Microplastic filter	Filters out anything that could disrupt the sample. And filters microplastics into a collection tank attached at the end of the filter. This filter was positioned under the ROV.	Was able to strongly attach the bottom of the ROV, and was able to navigate underwater while also collecting microplastic samples. The downside was that there was a lot of water collected along with the sample



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Size of the ROV	Constructed of PVC	The ROV's size allows for good navigation and speed without sinking under the weight itself.
Buoyancy	Constructed through pool noodles to help the ROV have a controllable submergence into the water. Assisted in taking weight off the ROV while navigating underwater.	Had to be adjusted multiple times to allow the ROV to be able to sink into the water, and that it would just float.

Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
There were no aspects of this that demanded.	



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